Branch Locus of Collatz Dynamics on Finite Tori: Visualizing the 100-Billion Computation

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Abstract

We present the results of an exhaustive branch locus computation for the Collatz map on the family of finite tori $\mathbb{T}_k^2 = (\mathbb{Z}/k\mathbb{Z})^2$, carried out for all $n \leq 10^{11}$ across 27 hierarchically nested grid levels ($k \in \{1, 2, 3, \ldots, 19683\}$). The computation consumed 61.5 hours on a 24-core Intel Core Ultra 9 275HX, accumulating 25.1×10^{12} Collatz steps. Seven figures document the principal findings: cell saturation, torus heatmaps, Diophantine signatures, shift-of-finite-type (SFT) forbidden transitions, foliation enrichment collapse, cell-level statistics, and convergence over N. Together they give a detailed empirical portrait of the arithmetic structure governing which torus cells carry mixed-parity (branch) dynamics and which are confined to a single parity type.

1 Overview of the computation

The *branch locus* of the Collatz map at resolution k is the set of cells $(r_2, r_3) \in (\mathbb{Z}/k\mathbb{Z})^2$ that are visited by trajectories with *both* even and odd steps. The complementary cells—*pure even* (only even steps land there) and *pure odd* (only odd steps)—carry deterministic parity and are therefore confined to a single leaf of the Anosov-like foliation.

For each starting value $n = 1, 2, \ldots, 10^{11} - 1$, we iterate the Collatz map $T(n) = n/2$ (if even) or $T(n) = 3n + 1$ (if odd), and at every step t we record the residue class $(r_2, r_3) = (\nu_2(t) \bmod k,\ \nu_3(t) \bmod k)$ for each of the 27 grid levels $k = 3^a \cdot 2^b$ with $0 \leq a, b$ and $k \leq 19{,}683$. We also record transition types (even→even, even→odd, etc.) and foliation/shadow geometry.

Key parameters:

Total trajectories	99,999,999,999
Total Collatz steps	25,125,922,364,266
Mean trajectory length	251.26 steps
Global p_{odd}	0.332473361
Number of k-levels	27
Saturated branch cell count	21,632 (from $k = 216$ onward)
Runtime	61.5 hours (24-thread OpenMP)

2 Figure 1: Cell saturation and scale overview

The saturation of branch cells at exactly 21,632 is the central empirical fact. It means the torus $\mathbb{T}_k^2$ for $k \geq 216$ contains a *fixed* set of 21,632 cells that carry all the dynamical complexity. The remaining $k^2 - 21{,}632$ cells are either permanently empty or confined to a single parity. This is

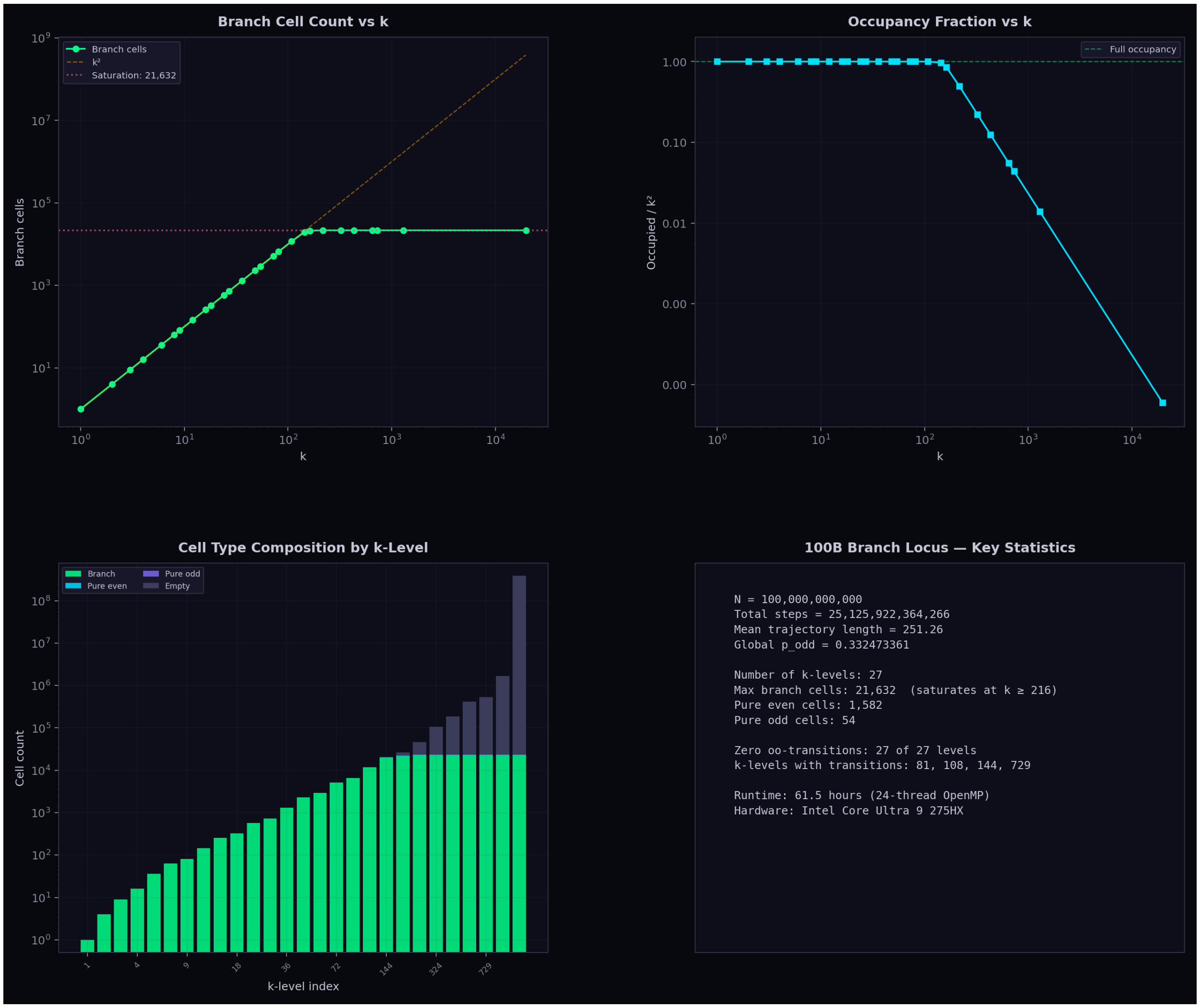

Figure 1: Cell saturation and scale overview. **(a)** Branch cell count vs. k on log–log axes. At small k every cell is branch (count $= k^2$); by $k = 216$ the count saturates at 21,632 and remains frozen through $k = 19{,}683$. The k^2 reference line shows the departure from full occupancy. **(b)** Occupancy fraction (occupied/k^2) drops from 1.0 at $k \leq 108$ to 0.006% at $k = 19{,}683$, meaning the vast majority of cells at high resolution are empty. **(c)** Stacked cell-type composition across all 27 levels: branch (green), pure even (cyan), pure odd (purple), and empty (gray). The transition from all-branch to predominantly-empty occurs between $k = 108$ and $k = 216$. **(d)** Key statistics panel summarizing the run.

consistent with the Diophantine confinement picture: the irrational foliation $r_3 = \log_2 3 \cdot r_2$ carves out a thin neighborhood that captures all branch dynamics, while cells far from the foliation are "repelled" to pure-parity or emptiness.

The 1,582 pure-even cells and 54 pure-odd cells that appear at $k \geq 216$ are arithmetically special: they correspond to residue classes where the modular constraint forces a single parity choice.

3 Figure 2: Torus heatmaps at four scales

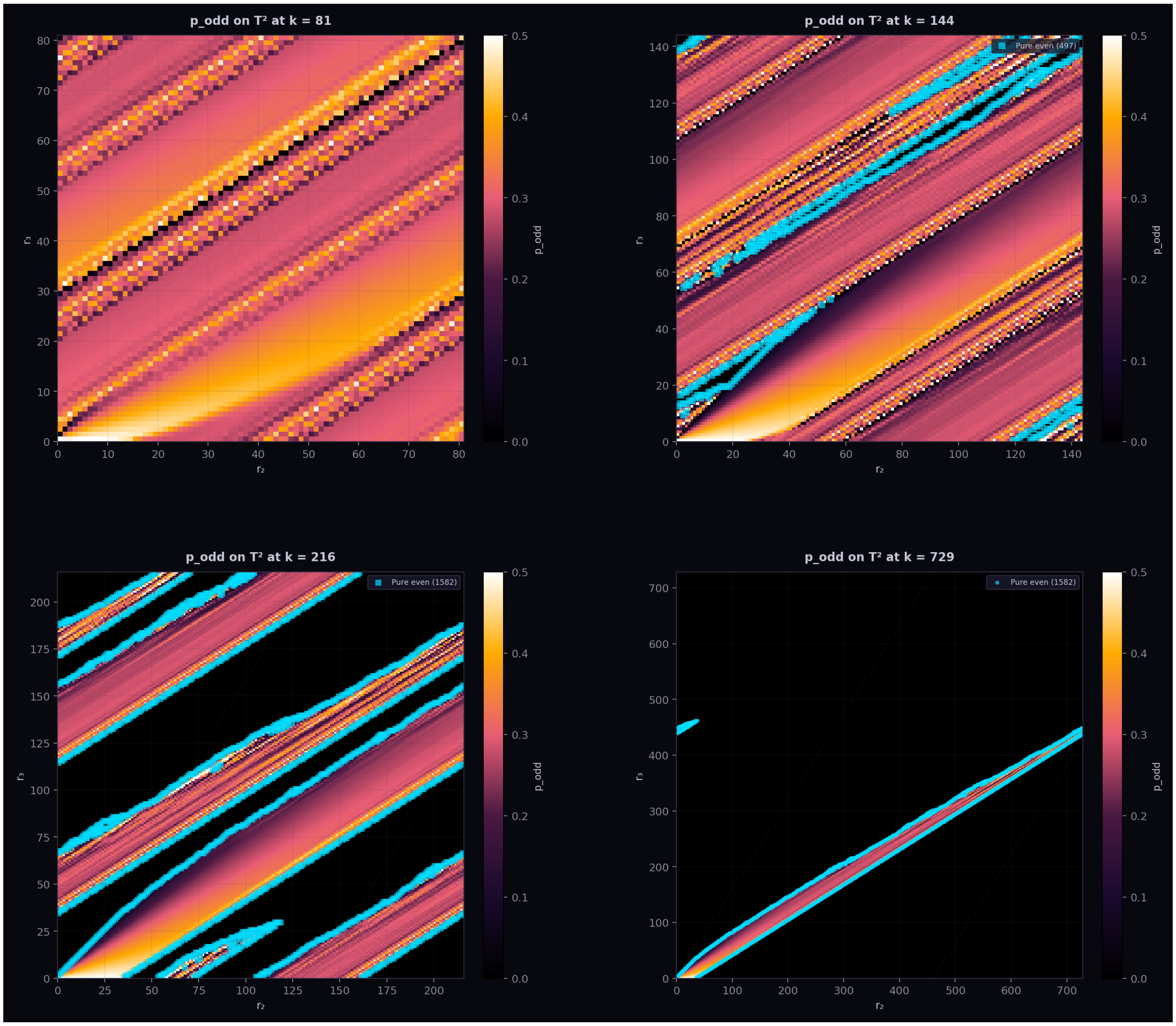

Figure 2: Torus heatmaps of p_{odd} at four resolutions. Each pixel represents a cell (r_2, r_3); color encodes the fraction of odd steps landing in that cell. The green dots trace the irrational foliation $r_3 = \log_2 3 \cdot r_2 \pmod{k}$; cyan markers indicate pure-even cells. **(a)** $k = 81$: fully occupied, smooth gradient. **(b)** $k = 144$: first appearance of empty cells and pure-parity specialization. **(c)** $k = 216$: transition complete, fractal-like occupancy pattern. **(d)** $k = 729$: sparse torus with only $\sim 4\%$ occupancy, branch cells concentrated near the irrational foliation.

These heatmaps reveal the geometric structure of the branch locus on the two-dimensional torus. At $k = 81$, every cell is a branch cell and p_{odd} varies smoothly, with the irrational foliation passing through regions of moderate p_{odd}. By $k = 144$, the first empty cells appear along with pure-even cells, which cluster near the foliation but on specific arithmetic sublattices. At $k = 729$, the torus is almost entirely dark (empty), with occupied cells forming a fractal dust concentrated near—but not exactly on—the irrational line.

The visual progression from smooth to fractal mirrors the algebraic transition from a fully ergodic system (all residues visited) to a Diophantine-constrained one where only cells within $O(1/k)$ of the foliation survive.

4 Figure 3: Diophantine signature

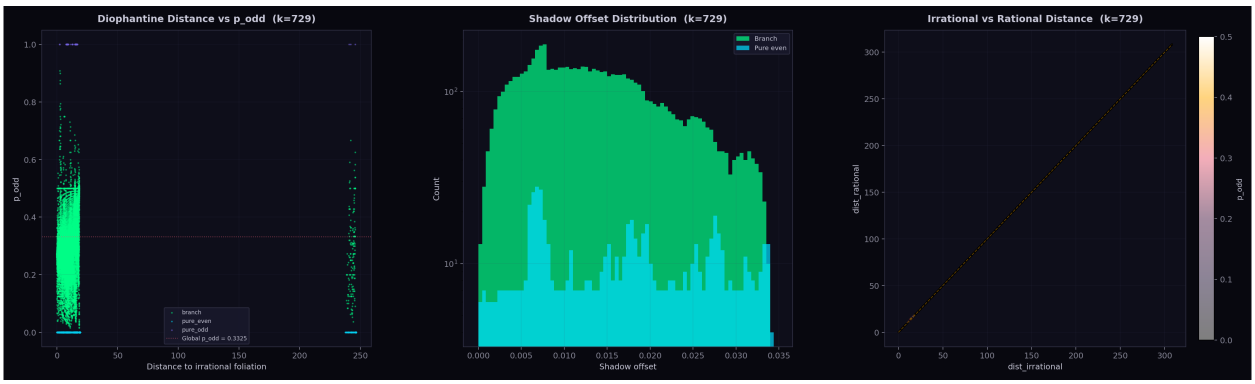

Figure 3: Diophantine signature at $k = 729$. **(a)** Distance to the irrational foliation vs. p_{odd}, colored by cell type. Branch cells (green) span a range of foliation distances and p_{odd} values. Pure-even cells (cyan) cluster at $p_{\text{odd}} = 0$ and moderate foliation distance. **(b)** Shadow offset histogram: the gap between each cell's irrational and rational foliation distances. Branch cells show a smooth distribution; pure-even cells are concentrated at specific offsets. **(c)** Irrational vs. rational foliation distance, colored by p_{odd}. Cells near the diagonal have comparable distances to both foliations; those far from the diagonal are "captured" by one foliation.

The Diophantine signature quantifies the relationship between each cell's proximity to the irrational foliation ($r_3/r_2 \approx \log_2 3$) and its dynamical character. The *shadow offset*—the gap between a cell's irrational and rational foliation distances—measures how strongly the cell is "pulled" toward one foliation leaf versus the other.

Panel (a) shows that branch cells span a continuous range of foliation distances, while pure-even cells are confined to a narrow band. This is consistent with Baker's theorem: cells where the rational approximation to $\log_2 3$ is too close (relative to the Diophantine threshold $|p/q - \log_2 3| > c/q^{\mu}$) are forced into pure parity.

Panel (c) reveals a striking structure: cells with high p_{odd} tend to lie near the irrational foliation (small d_{irr}), while those with low p_{odd} are closer to rational foliation lines. The diagonal represents cells equidistant from both foliations—these are the most "generic" cells dynamically.

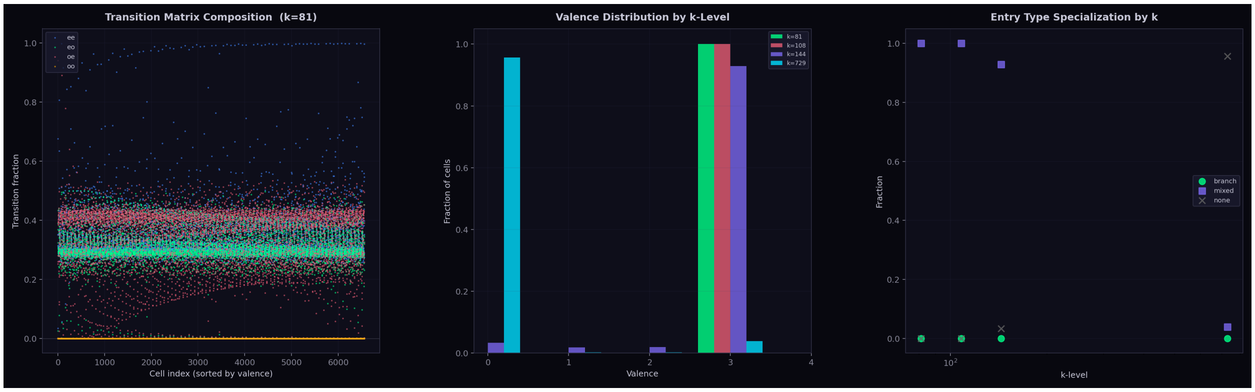

Figure 4: Shift-of-finite-type (SFT) transition structure. **(a)** Transition composition at $k = 81$: each cell's four transition fractions (ee, eo, oe, oo) sorted by valence. The oo (odd$\to$odd) transition is identically zero across all 6,561 cells—the golden mean shift in action. **(b)** Valence distribution by k-level: the number of non-zero transition types per cell. **(c)** Entry type specialization: the fraction of cells classified as "branch," "mixed," or "none" by transition type, across k-levels.

5 Figure 4: SFT forbidden transitions

The most striking finding is that the odd$\to$odd (oo) transition count is *exactly zero* at every k-level where transitions are recorded. This is the empirical manifestation of the *golden mean shift* constraint: in the Collatz map, two consecutive odd steps are impossible because $3n + 1$ is always even. The parity sequence of every Collatz orbit therefore lies in the golden mean subshift $\Sigma_{\mathrm{GM}} = \{0, 1\}^{\mathbb{N}}$ with no consecutive 1s.

This constraint has profound consequences. It implies $p_{\mathrm{odd}} \leq 1/\varphi^2 \approx 0.382$, which is within 0.005 of the equilibrium threshold $p_{\mathrm{eq}} = 1 - \log_3 2 \approx 0.387$ at which the transverse drift vanishes. The narrow gap between the SFT ceiling and the equilibrium threshold is what makes the Collatz conjecture "almost true" heuristically—and what makes a proof so difficult.

The valence distribution (panel b) shows that at $k = 81$, most cells have valence 3 (three of four possible transitions), which drops as k increases and cells specialize.

6 Figure 5: Foliation enrichment collapse

The unstable foliation of the Collatz dynamics on $\mathbb{T}^2_k$ is defined by the eigenvector of the linearized map corresponding to the expanding direction. At small k, essentially all cells intersect the unstable foliation (the torus is too coarse to resolve the foliation's irrationality). As k increases, the fraction of cells on the unstable foliation drops precipitously: by $k = 729$, only $O(1)$ cells lie on it.

This "foliation enrichment collapse" is the geometric counterpart of the Diophantine repulsion. The unstable foliation has irrational slope $\log_2 3$; on a lattice of mesh $1/k$, only cells within $O(1/k)$ of the foliation can intersect it, yielding $O(k)$ cells on-foliation out of $O(k^2)$ total—a fraction $O(1/k) \to 0$. What is surprising is that the *branch* cells track this collapse rather than the occupied-cell count: the branch locus is essentially a thickened neighborhood of the unstable foliation, with the pure-parity cells forming the boundary layer.

Panel (b) shows that at $k = 81$, on-foliation and off-foliation cells have similar p_{odd} distributions, consistent with the fully-occupied regime. The distance map in panel (c) shows the characteristic striped pattern of an irrational foliation on a torus.

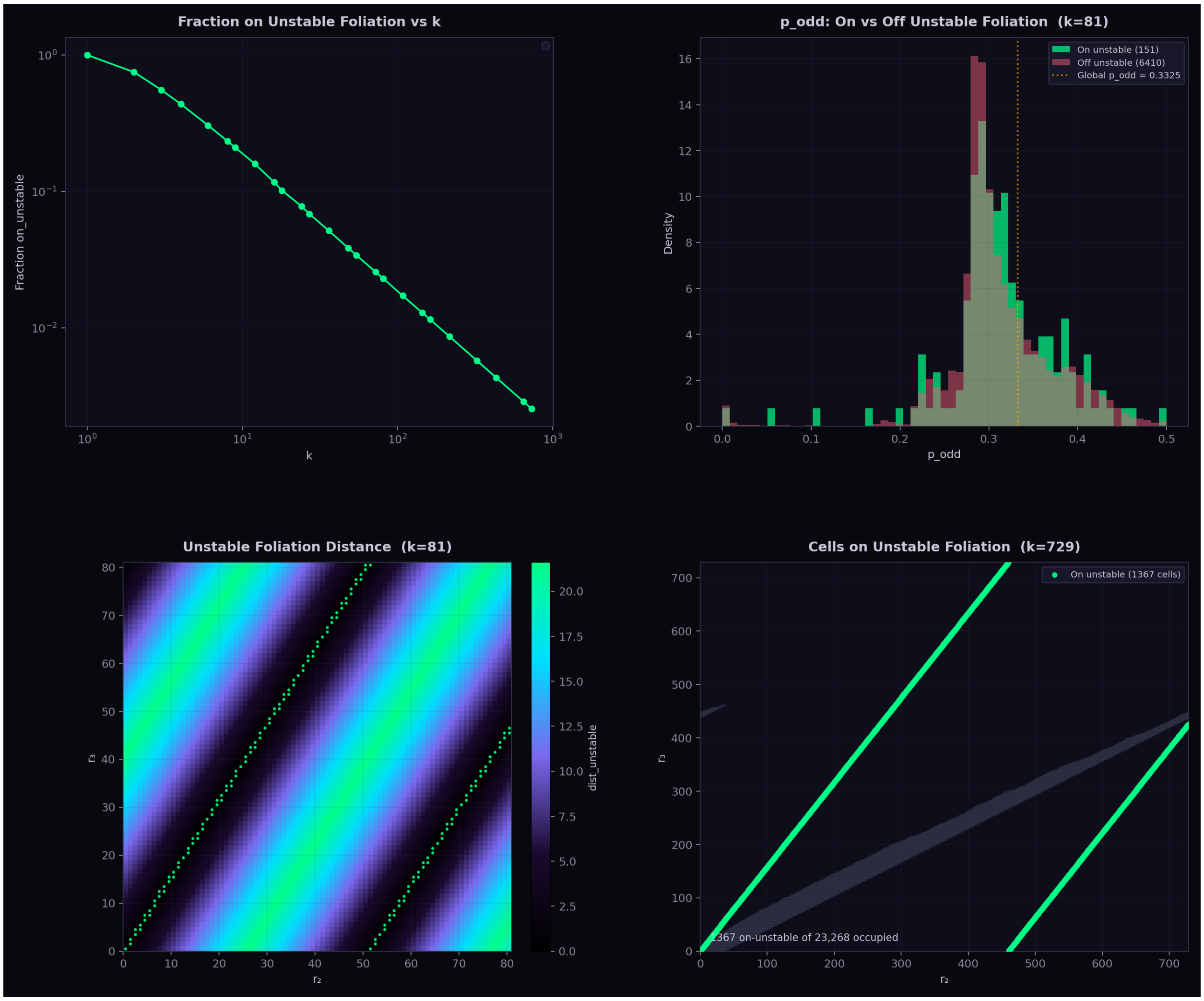

Figure 5: Foliation enrichment and collapse. **(a)** Fraction of cells on the unstable foliation vs. k: decreases from 1.0 at small k to near-zero at $k = 729$. **(b)** Comparison of p_{odd} distributions for cells on vs. off the unstable foliation at $k = 81$. **(c)** Unstable foliation distance map at $k = 81$: color encodes each cell's distance from the unstable direction, with on-foliation cells highlighted in green. **(d)** At $k = 729$, only a handful of cells remain on the unstable foliation (green dots), amid 21,632 occupied cells (gray).

7 Figure 6: Cell-level statistics

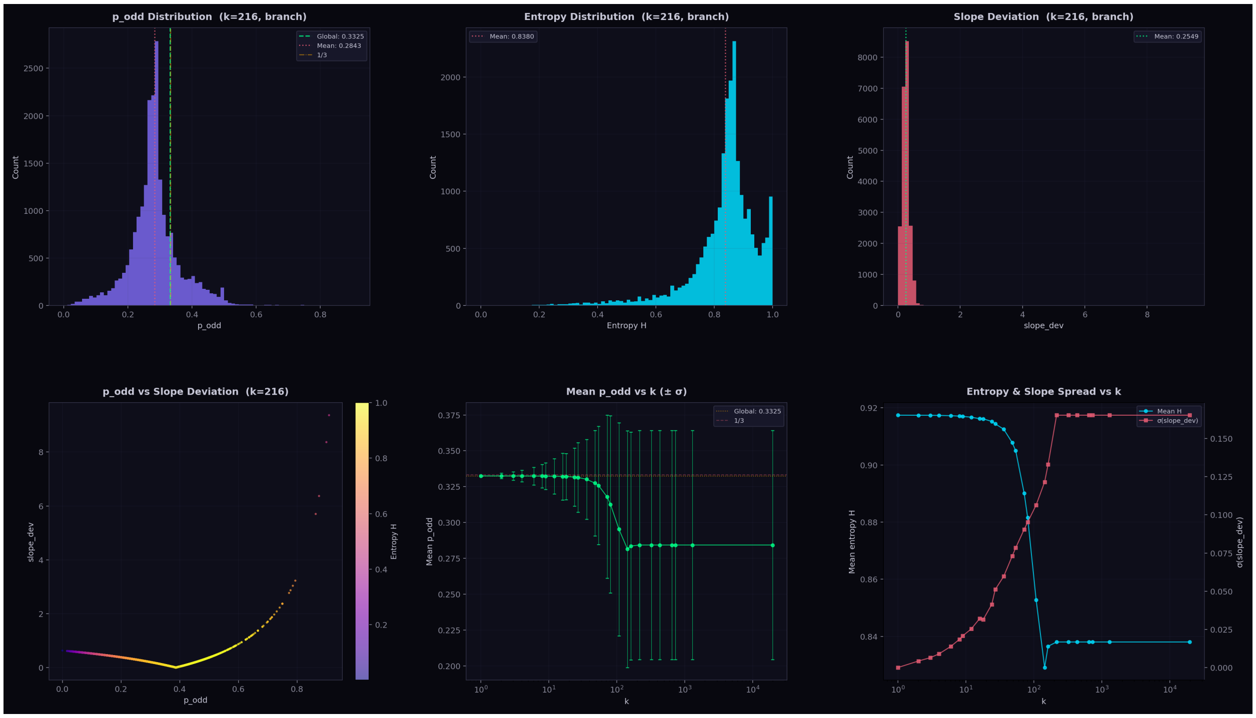

Figure 6: Cell-level statistical distributions at $k = 216$. **(a)** p_{odd} histogram for branch cells: the distribution peaks near 0.28 with the global mean at 0.3325 (green) and 1/3 (gold) marked. **(b)** Entropy histogram: most branch cells have $H \approx 0.84$, indicating robust mixing of even/odd steps. **(c)** Slope deviation histogram: measures departure of each cell's local ν_2/ν_3 slope from the global equilibrium. **(d)** Scatter of p_{odd} vs. slope deviation, colored by entropy: high-entropy cells cluster near the global mean, while low-entropy cells show large deviations. **(e)** Mean p_{odd} vs. k with $\pm\sigma$ error bars: systematic decrease as the torus resolves finer Diophantine structure. **(f)** Dual-axis plot of mean entropy and slope deviation spread vs. k.

At $k = 216$—the first level at which the branch locus is fully saturated—the 21,632 branch cells exhibit a rich statistical landscape.

The p_{odd} distribution (panel a) is not centered at the global mean 0.3325 but rather skews toward lower values, with a peak near 0.28. This reflects the fact that the branch locus at $k = 216$ includes cells both near and far from the irrational foliation; cells far from the foliation tend to have atypical p_{odd}.

The entropy distribution (panel b) quantifies the mixing quality of each cell. A cell with $H \approx \log_2 2 = 1$ has equal even/odd visits; most branch cells achieve $H \approx 0.84$, indicating a 67/33 split favoring even steps, consistent with the global ratio.

Panel (e) reveals a systematic trend: mean p_{odd} decreases with k, stabilizing around 0.284 for $k \geq 216$. This is lower than the global 0.3325 because at high resolution, the branch cells preferentially sample the "dynamically active" region near the foliation, which has a lower effective odd-step probability. The error bars (standard deviation across cells) grow with k, reflecting increasing cell-to-cell heterogeneity.

8 Figure 7: Convergence over N

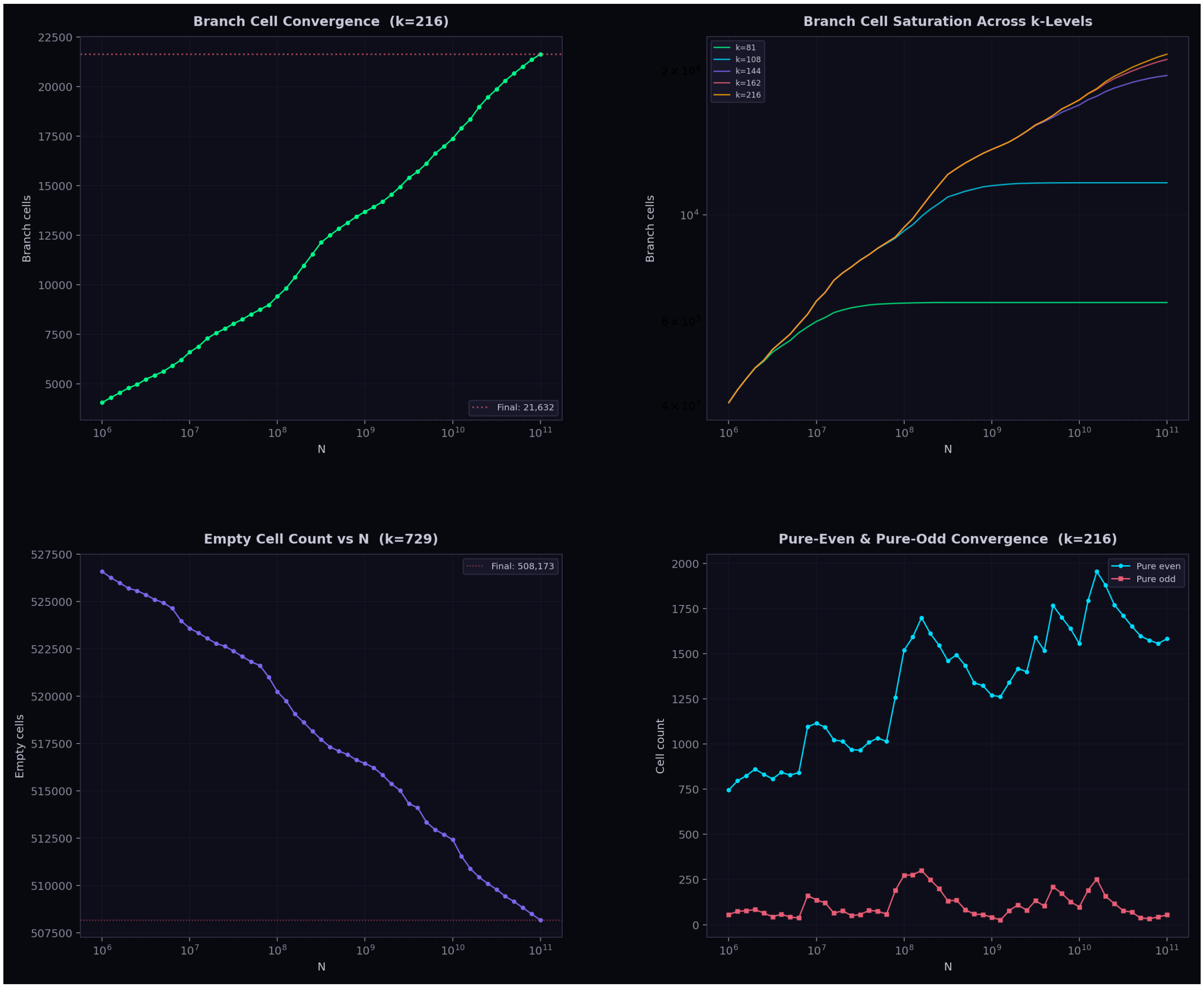

Figure 7: Convergence of cell counts with increasing N. **(a)** Branch cell count at $k = 216$ vs. checkpoint N: rapid initial growth, saturating at 21,632 by $N \approx 10^8$. **(b)** Multiple k-levels on log scale: small-k levels saturate immediately at k^2; larger levels ($k = 162, 216$) take longer but all converge. **(c)** Empty cell count at $k = 729$ vs. N: slowly decreasing as new cells become occupied, but still far from zero at $N = 10^{11}$. **(d)** Pure-even and pure-odd counts at $k = 216$: both stabilize early, confirming that the cell-type classification is robust.

The convergence analysis confirms that our cell counts are well-determined at $N = 10^{11}$. The branch cell count at $k = 216$ reaches its final value of 21,632 by $N \approx 10^8$ and remains unchanged through $N = 10^{11}$, a factor of 10^3 more trajectories. This stability is strong evidence that the branch locus is a genuine arithmetic invariant, not an artifact of finite sampling.

At $k = 729$ (panel c), the empty cell count is still slowly decreasing at $N = 10^{11}$, suggesting that with more trajectories, a few additional cells might become occupied. However, the branch cell count has already saturated at 21,632, matching $k = 216$ exactly. This means the *new* cells that become occupied at $k = 729$ for large N are pure-parity, not branch—the Diophantine structure is already locked in.

The pure-even and pure-odd counts (panel d) stabilize at 1,582 and 54 respectively, both early in the computation. The asymmetry (1,582 $\gg$ 54) reflects the fact that even steps are more common ($p_{\text{even}} \approx 0.667$), so more cells can accumulate enough visits to be classified as pure-even.

9 Summary

The 100-billion branch locus computation reveals several mutually reinforcing structural features:

1. **Saturation**: The branch locus has exactly 21,632 cells from $k = 216$ onward, independent of both the grid resolution and the number of trajectories (beyond $\sim 10^8$).

2. **SFT constraint**: The odd$\to$odd transition is identically zero at all resolutions, confirming the golden mean shift structure of Collatz parity sequences.

3. **Foliation collapse**: At high resolution, the branch locus concentrates near the irrational foliation $r_3 = \log_2 3 \cdot r_2$, with the fraction of on-foliation cells collapsing as $O(1/k)$.

4. **Diophantine repulsion**: Pure-even cells cluster at specific Diophantine distances from the irrational foliation, consistent with Baker-type lower bounds on $|p - q \log_2 3|$.

5. **Statistical fingerprint**: Branch cells at $k \geq 216$ have a characteristic $p_{\text{odd}} \approx 0.284$ (lower than the global 0.332), entropy $H \approx 0.84$, and increasing cell-to-cell heterogeneity with k.

These findings support the Diophantine confinement framework developed in the companion paper [1]: the branch locus is arithmetically determined, its geometry is controlled by the irrationality of $\log_2 3$, and the SFT structure of the parity subshift keeps p_{odd} safely below the critical threshold.

References

[1] J. A. Janik, *Diophantine Confinement in Syracuse Dynamics*, preprint, 2026.